

Artificial Intelligence for Video Games

Chasing Flock

Project due Feb. 5, 2025

Goal

The goal of this project is to simulate a flock of agents moving inside a scene characterized by moving obstacles. Moreover, a target is placed randomly in the scene: the path to reach the target must be computed, and the flock must move towards the target.

Setup

For simplicity, the project can consider a 2D closed environment (a 100 by 100 meters square area). The sides of the square must be considered as walls for the flock.

Several obstacles (graphically, grey circles of radius = 1 meter) are placed randomly in the scene. They must move back and forth horizontally inside the scene (without colliding with each other). The speed of every obstacle must be between 5 and 20 meters/s.

A random target (graphically, a small red circle of radius = 0.1 m) is placed randomly in the scene. At the beginning of the simulation, a corner of the scene is picked up randomly, and the target is placed randomly at a maximum distance of 10 meters from the corner.

A flock of 50 agents moves inside the scene, each agent with a fixed speed of 10 meters/s. The flock follows the original approach of Craig Reynolds (as seen during the course). However, the flocking movement is affected by the presence of the target: the path to reach the target must be computed, and the flock will move towards the target, taking into consideration the moving obstacles.

When one of the agents reaches the target, a new target is placed randomly in the scene (following the same approach used for the first target but excluding the current corner in the computation of the position of the new target), and the flock must move towards the new target.

Constraints

The obstacles must be between 10 and 30, the number can be changed using the GUI.

Movement must be calculated using a kinematics approach. I.e., Unity physics subsystem cannot be involved in the computation of the final position of each agent.